NAU
NORTHERN ARIZONA UNIVERSITY

Hydro-Power Collegiate Competition (HCC) Capstone

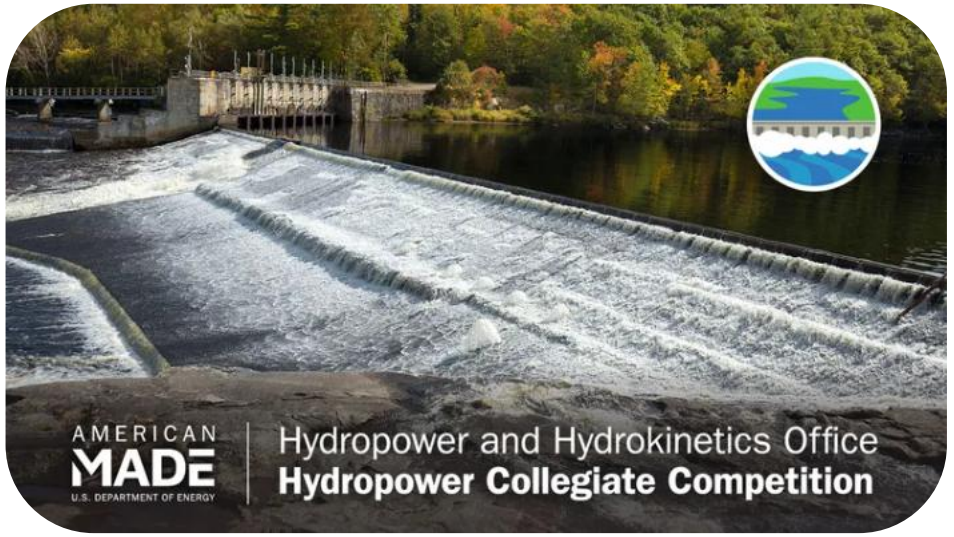
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Project Description

Select and evaluate a U.S. non-powered dam (NPD) and develop a feasible hydropower retrofit that increases renewable-energy generation while maintaining the dam's existing function.

Goal: Generate renewable electricity while maintaining the dam's original purpose.

Siting
↓
Design
↓
Outreach
↓
Presentation
↓
Build

Advisor:
Carson Pete

Sponsor:
NAU Energy Club

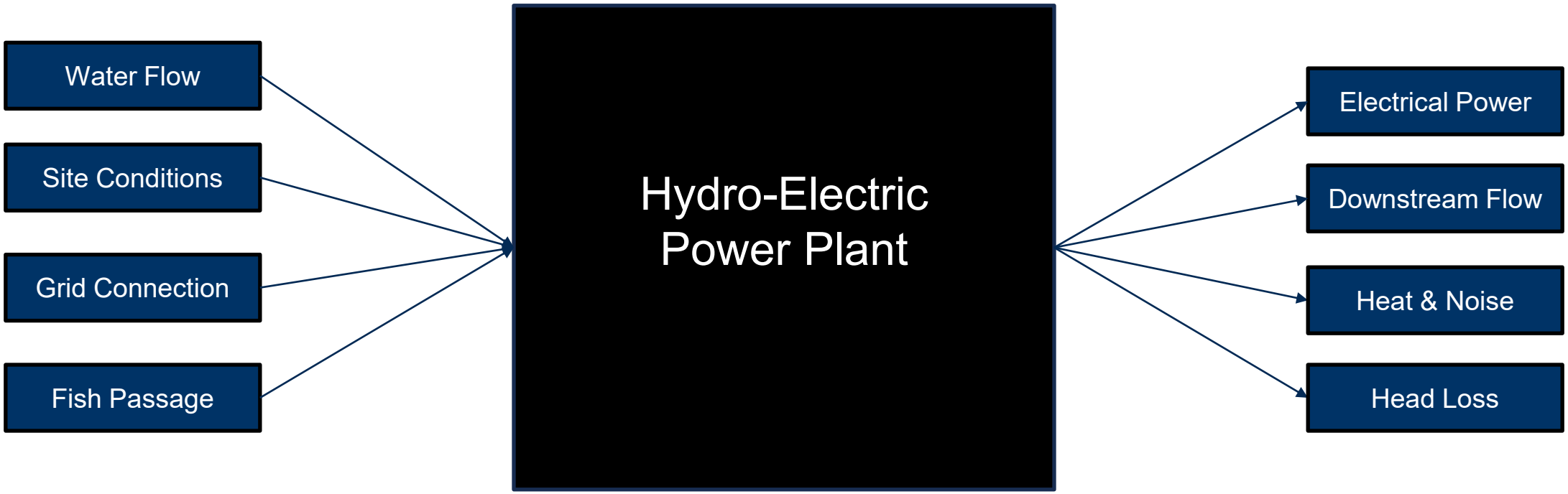
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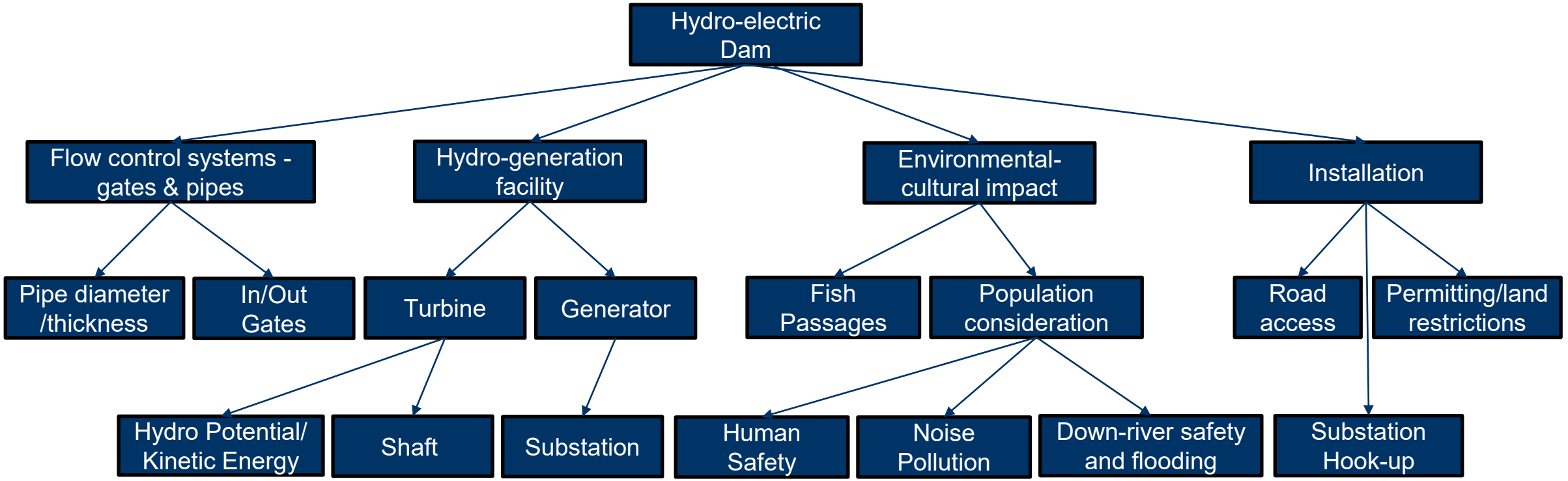
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Loggins

Black Box


Water Flow
Site Conditions
Grid Connection
Fish Passage
Hydro-Electric Power Plant
Electrical Power
Downstream Flow
Heat & Noise
Head Loss

Functional Decomposition



Concept Generation

Current Elimination Criteria:
- Generation Capacity between 1 and 10 MW

Soon to Come:
- No national parks or protected lands
- No possible fish passages

Soft Elimination (in development):
- Geographical concerns (mountains, valleys) for transmission lines
- Proximity to residential areas

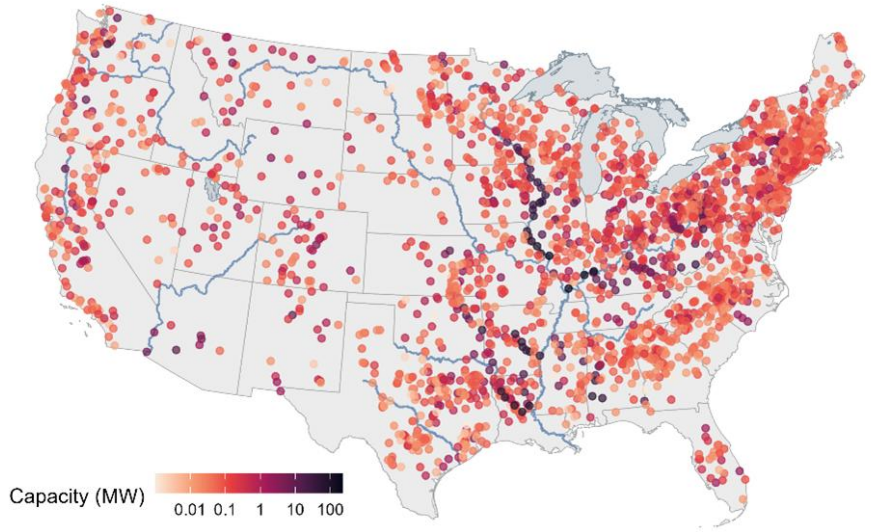
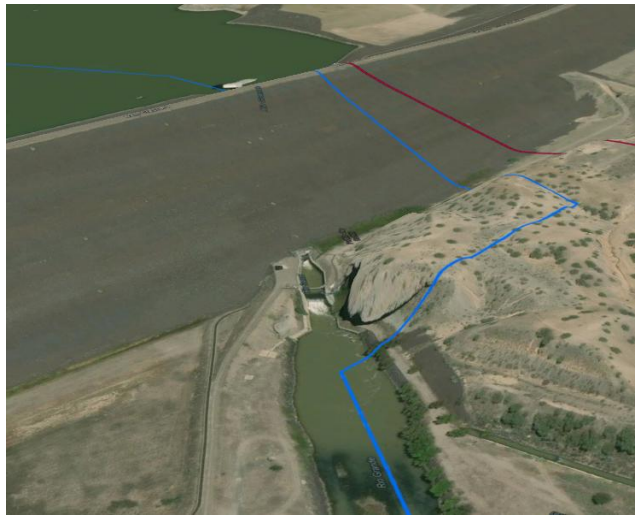


Image of potential hydropower conversions [21]

Concept Generation

	FISH BARRIER DAM	NEW CROTON RESERVIOR DAM	COCHITI DAM
			
Capacity (MW)	2.56 [21]	5.06 [21]	9.91 [21]
Output (MWh/yr)	19868 [21]	20192 [21]	58407 [21]
Average Flow (m^3/s) & CV	15.57 / 0.02 [21]	11.48 / 0.64 [21]	31.98 / 0.61 [21]
Head (m) & CV	18.23 / 0.04 [21]	34.84 / 0.04 [21]	33.43 / 0.02 [21]
Distance to Substation (km)	2.09 [22]	0.32 [22]	0.32 [22]
Utility Energy Cost ($/kWh)	0.31 [23]	0.29 [23]	0.13 [23]
Penstock (m)	2.7	2.2	3.7

Concept Generation

Turbine Type	Design Head	Design Flow	Expected Relevance
Pelton	High	Low	Low
Francis	Medium	Medium	High
Kaplan	Low	High	High

Turbine concepts are generated from site specific head and flow conditions

Additional considerations for future:

- Efficiency
- Environmental constraints
- Constructability

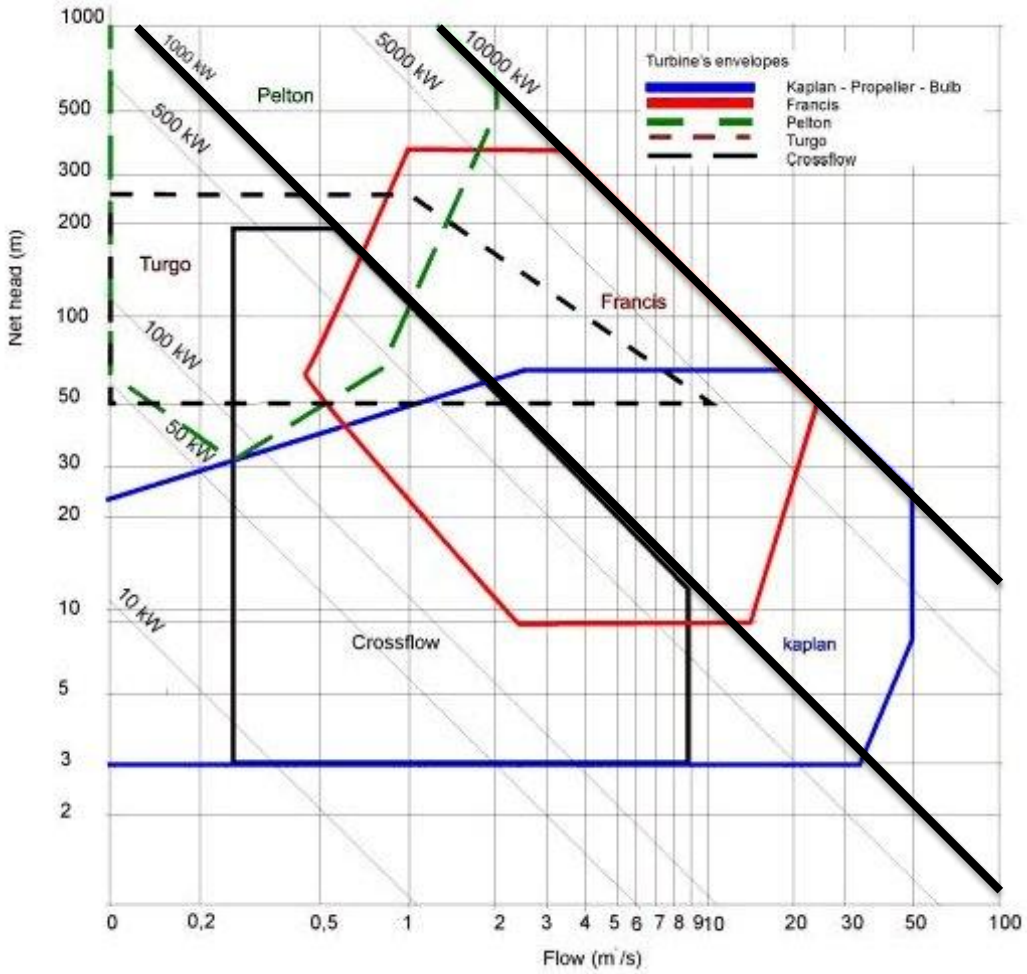
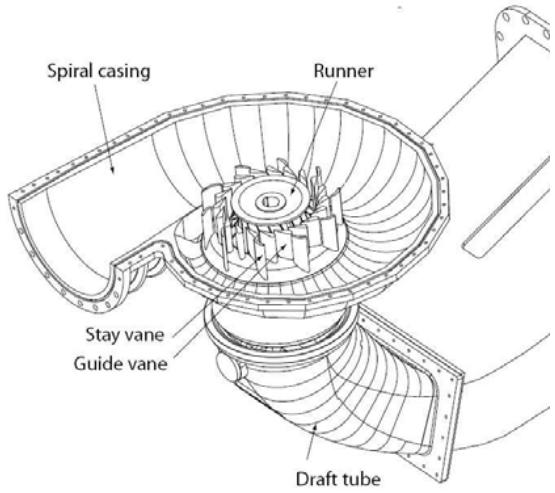
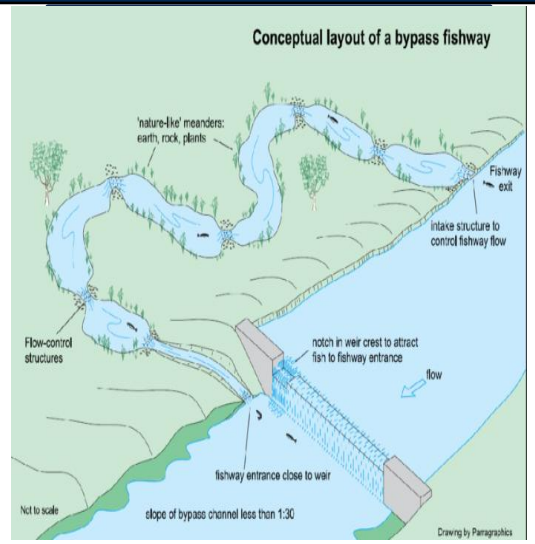
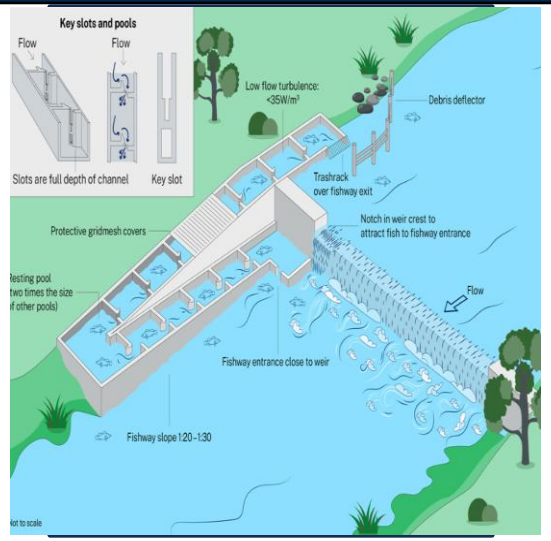


Image of flow and head acceptance criteria of different turbines [24]

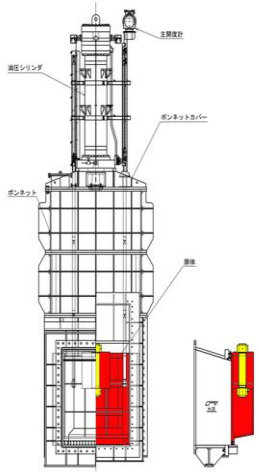
Concept Generation

	FRANCIS TURBINE	PELTON TURBINE	KAPLAN TURBINE
			
Type	Reaction: fully submerged runner driven by water pressure [12]	Impulse: a water jet strikes buckets in open air [12]	Reaction: fully submerged propeller-style runner [12]
Head	25-350 m [12],[13]	50-1300 m [12],[13]	2-40 m [12],[13]
Flow rate	Medium , 0.45- 25 m^3/s [12]	Low 0.02-2.5 m^3/s [12]	High 0.3-50 m^3/s [12]
Peak efficiency	0.94 [12]	0.89-0.90 [12]	0.93 [12]
Minimum operating flow	50% of design flow [12],[13]	10% of design flow [12],[13]	15% of design flow [12],[13]
Specific speed (nQHg)	0.05-0.33 [13]	0.005–0.025 [13]	0.19–1.55 [13]

Concept Generation- Fish Passage

	NATURE-LIKE BYPASS	FISH LIFT	VERTICAL-SLOT FISHWAY
			
Advantages	-Can act as additional habitat [19]. -Minimal concrete and construction cost [20].	-Preferred choice for dams over 25 ft [20].	-Strong attraction 80% of fish find passage [19]. -Works well with wide slots and low slopes/lower dams [20].
Disadvantages	-Low attraction rate 48% [19]. -Requires very low slopes [20]. -Performance varies by site [20].	-Harder to attract fish to lift [20]. -Not consistent operation [20].	-Low passage once inside 45% [19] -Narrow slots, single file movement [20].

Concept Generation – Penstock Gate

	Fixed Wheel Vertical Lift	Tractor Gate	High Pressure Hydraulic Slide Gate
			 
Advantages	-Suitable for large openings -Gravity closure for emergency -Proven successful for emergencies	-Even loading along guides -Good for heavy loaded openings -Used as emergency stops and regular closures	-Good for closing against flow and uneven head -Not as many moving parts -Positive hydraulic actuation
Disadvantages	-Need a lot of space -Large guides -Needs maintenance -Debris can affect seal	-Many moving parts -Needs proper maintenance access -Prone to corrosion	-Sliding friction means more wear on seals -Not a good option for large penstock openings.

Engineering Calculations

QUESTION

What turbine size (design flow) gives the best balance of energy and cost at the selected dam, and how reliably can the river supply it?

METHOD

-Built a flow duration curve [6] from 30 years of USGS annual mean flows [8].
-Selected a turbine from the head of the site [12], [13], and calculated the short-cut design flow (Q_nom) [1].
-Tested a range of design flows with the power formula [12] and ORNL cost equations [2] to find the lowest cost of energy [18].

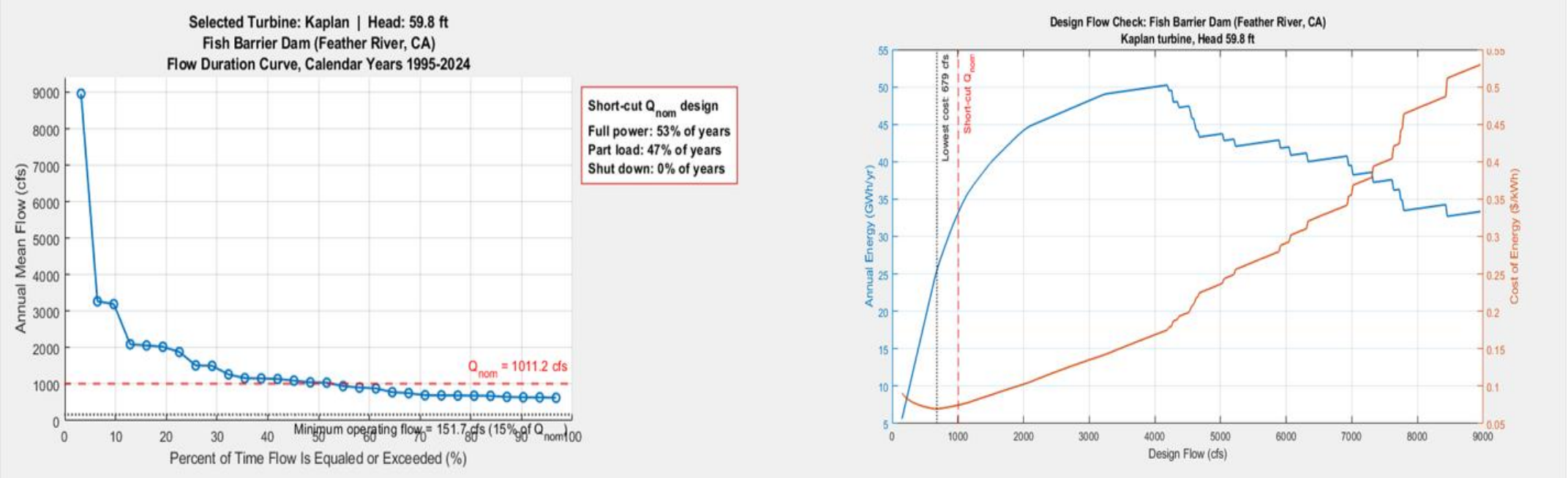
Flow Duration Curve and Design Flow		
Exceedance	$P = \frac{m}{n+1} \times 100\%$	[6]
Short-cut design flow	$Q_n = \left[\frac{\gamma q_{50}}{1+(\gamma-1)q_{50}}\left(1-\frac{q_{min}}{q_{max}}\right)+\frac{q_{min}}{q_{max}}\right]Q_{max}$	[1]
Normalized flows	$q_{50} = \frac{Q_{50}}{Q_{max}}, \quad q_{min} = \frac{Q_{min}}{Q_{max}}$	[1]
Min. operating flow	$Q_{min,\,op} = 0.15\,Q_n$ (Kaplan), $\quad 0.50\,Q_n$ (Francis)	[12], [13]

Design Flow Check: Energy and Cost		
Power	$P = \frac{Q\,H\,\eta}{11.8}$	[12]
Annual energy	$E = Q_{used}\frac{H\,\eta}{11.8} \times 8760$	[12]
Capital cost	$ICC = 11{,}489{,}245\,P^{0.976}H^{-0.240}$	[2]
Annual O&M	$C_{OM} = 225{,}417\,P^{0.547}$	[2]
Capital recovery	$CRF = \frac{i(1+i)^n}{(1+i)^n-1}$	[18]
Cost of energy	$LCOE = \frac{ICC \cdot CRF + C_{OM}}{E}$	[18]

Assumptions		
Plant efficiency	$\eta = 0.85$ (overall turbine, generator and losses)	assumed
Project life	$n = 30$ years (payment period for capital cost)	assumed
Discount rate	$i = 7\%$ per year	assumed
Inflation	2% per year, costs brought to 2026 dollars	assumed
Turbine coefficient	$\gamma = 0.364$ (Kaplan), 0.422 (Francis)	[1]
Flow data	30 years of annual mean flows (screening level)	[8]

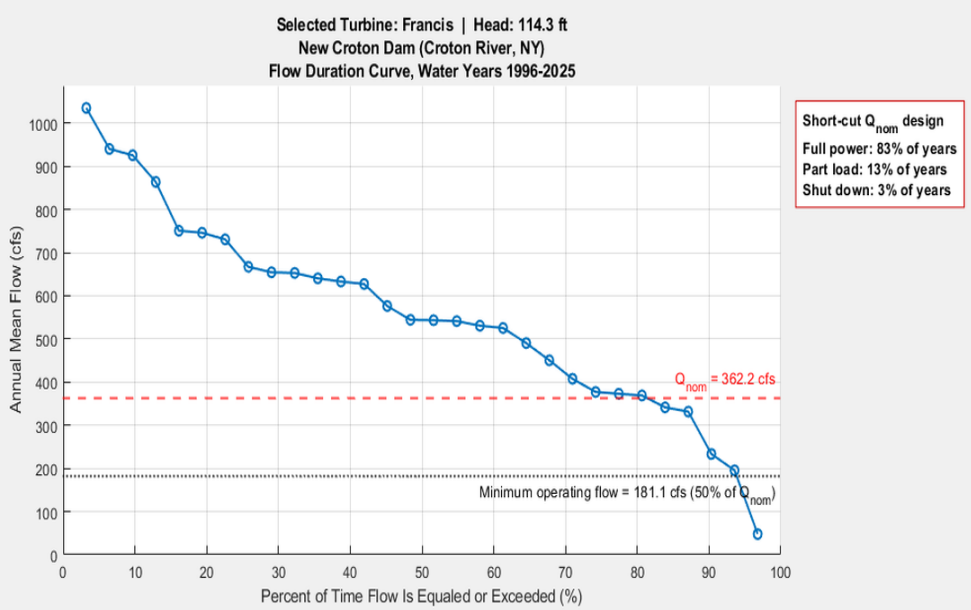
Engineering Calculations

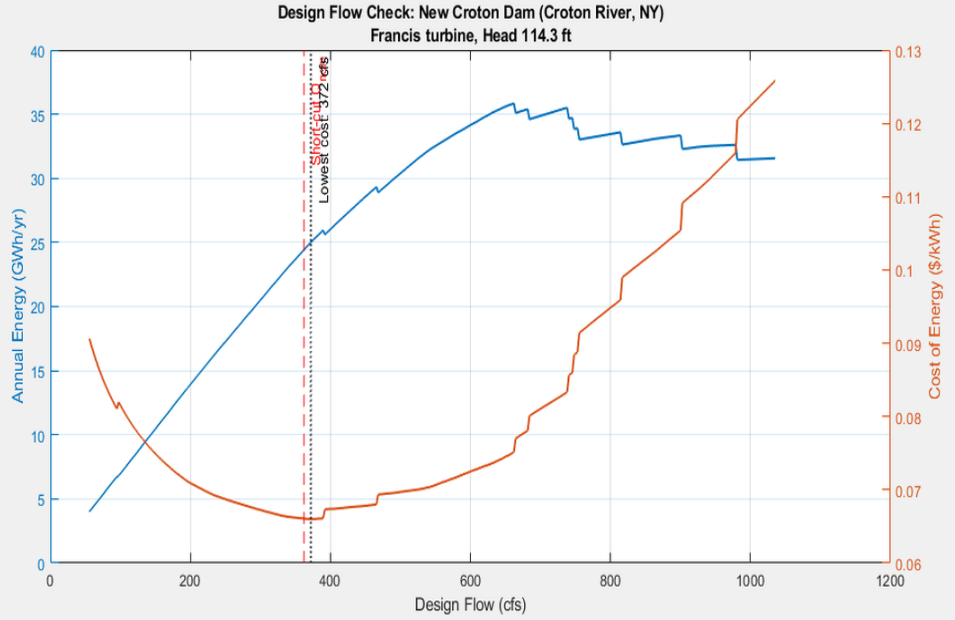
About 679 cfs (~2.9 MW) gives the lowest cost of energy



Engineering Calculations

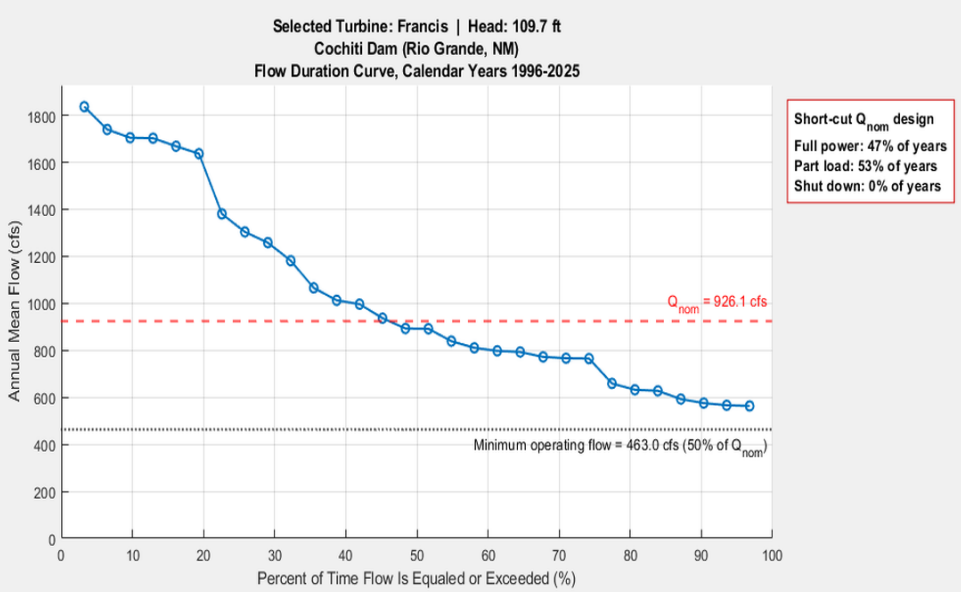
About 372 cfs (~3.1 MW) gives the lowest cost of energy, almost exactly matching the short-cut design flow of 362 cfs, which confirms the method at this site.

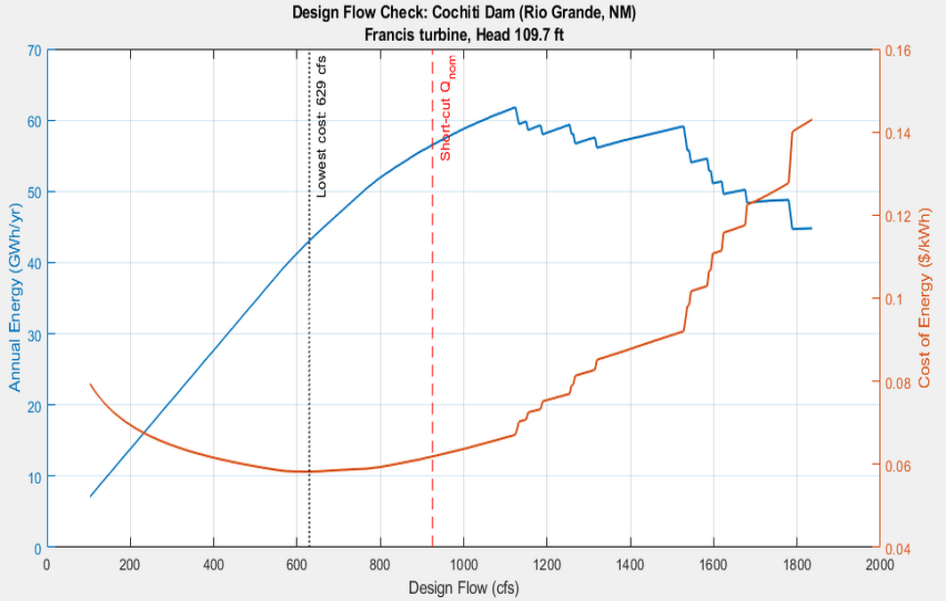




Engineering Calculations

About 629 cfs (~5.0 MW) gives the lowest cost of energy





Engineering Calculations

Results-

This table shows the turbine design for each dam that produces the lowest cost of energy. The listed capacity is the amount of power available at that most economical operating point.

Results: Best Design Flow (Lowest Cost of Energy)			
Dam	**Turbine**	**Installed Capacity (MW)**	**Cost of Energy ($/kWh)**
Fish Barrier	Kaplan	2.9	$0.070
New Croton	Francis	3.1	$0.066
Cochiti	Francis	5.0	$0.058

Engineering Calculations

Design Question

How does penstock diameter affect water velocity and hydraulic head loss in consideration for system effieceny?

Given Values and Assumption

Parameter	Fish Barrier	New Croton	Cochiti
Design Flow, Q (m^3/s)	19.23	10.53	17.81
Penstock Diameter, D (m)	2.7	2.2	3.7
Gross Head, H_g (m)	18.23	34.84	33.43
Assumed Penstock Length, L (m)	100	100	100
Assumed Darcy Friction Factor, f	0.020	0.020	0.020
Assumed Minor Loss Coefficient, ΣK	1.5	1.5	1.5

Engineering Calculations

Equation Name	Equation Format	Meaning	Fish Barrier	New Croton	Cochiti
Cross-Sectional Area	$A = \frac{\pi D^2}{4}$	Area available for water flow	5.726 m²	3.801 m²	10.752 m²
Water Velocity	$V = \frac{Q}{A}$	Average velocity through penstock	3.359 m/s	2.770 m/s	1.656 m/s
Friction Head Loss	$h_f = f \frac{L}{D} \frac{V^2}{2g}$	Energy lost due to pipe friction	0.426 m	0.356 m	0.076 m
Minor Head Loss	$h_m = \Sigma K \frac{V^2}{2g}$	Losses from entrances, bends, and fittings	0.862 m	0.587 m	0.210 m
Total Head Loss	$h_L = h_f + h_m$	Combined hydraulic losses	1.288 m	0.942 m	0.285 m
Net Head	$H_{net} = H_g - h_L$	Head remaining for turbine operation	16.942 m	33.898 m	33.145 m
Head Loss Percentage	$\frac{h_L}{H_g} \times 100$	Percentage of gross head lost	7.07%	2.70%	0.85%

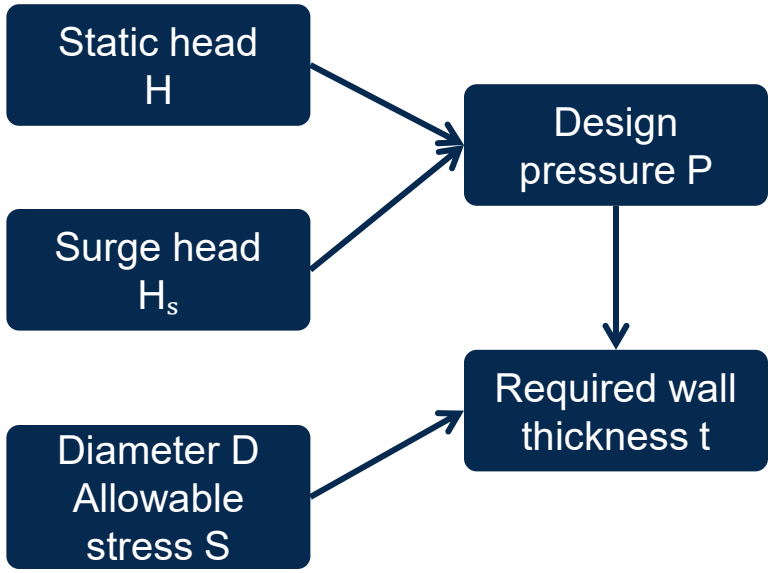
Engineering Calculations

Design Question

Design Question: What pipe wall thickness is required to carry the specified water flow and withstand internal pressure from head and surge?

Governing Equations

Concept	Equation	Meaning
Diameter and velocity	$D = \sqrt{(4A/\pi)}$ $v = Q/A$	Pipe diameter and water speed
Design pressure	$P = \rho g(H + H_s)$	Static and surge pressure
Allowable hoop stress	$S = S_y/n$	Material strength with safety factor
Thin-wall hoop stress	$\sigma_h \approx PD/(2t)$	Tensile stress around pipe wall
Pressure-only thickness	$t_p = (D/2) \times [\sqrt{((S+P)/(S-P))} - 1]$	Lamé sizing for internal pressure
Nominal wall thickness	$t_n = (t_p + c)/(1 - m)$	Corrosion and mill allowance



Q: flow rate A: internal flow area
ρ: water density g: gravity
S_y: tensile yield strength n: safety factor
c: corrosion allowance m: tolerance fraction

Engineering Calculations

Equation Name	Equation Format	Meaning	Fish Barrier	New Croton	Cochiti
Cross-Sectional Area	$A = \pi D^2/4$	Internal area available for water flow	5.726 m²	3.801 m²	10.752 m²
Inside Diameter	$D = \sqrt{(4A/\pi)}$	Confirmed inside diameter of the pipe	2.700 m	2.200 m	3.700 m
Water Velocity	$v = Q/A$	Average water speed through the pipe	2.719 m/s	3.020 m/s	2.974 m/s
Design Pressure	$P = \rho g(H + H_s)$	Pressure from head with H_s = 0	0.1788 MPa	0.3418 MPa	0.3279 MPa
Allowable Hoop Stress	$S = S_Y/n$	205 MPa yield strength divided by 1.2	170.83 MPa	170.83 MPa	170.83 MPa
Thin-Wall Hoop Stress	$\sigma_h \approx PD/(2t_p)$	Approximate stress at calculated t_p	170.74 MPa	170.66 MPa	170.67 MPa
Pressure-Only Thickness	$t_p = (D/2) \times [\sqrt{((S+P)/(S-P))} - 1]$	Lamé minimum wall for internal pressure	1.414 mm	2.203 mm	3.555 mm
Nominal Wall Thickness	$t_n = (t_p + c)/(1 - m)$	Same as t_p when c = 0 and m = 0	1.414 mm	2.203 mm	3.555 mm

Inputs: ρ = 1000 kg/m³, g = 9.81 m/s², S_Y = 205 MPa, n = 1.2, H_s = 0, c = 0, m = 0.
Static pressure sizing only. These calculated walls exclude buckling, joints, support loads and surge.

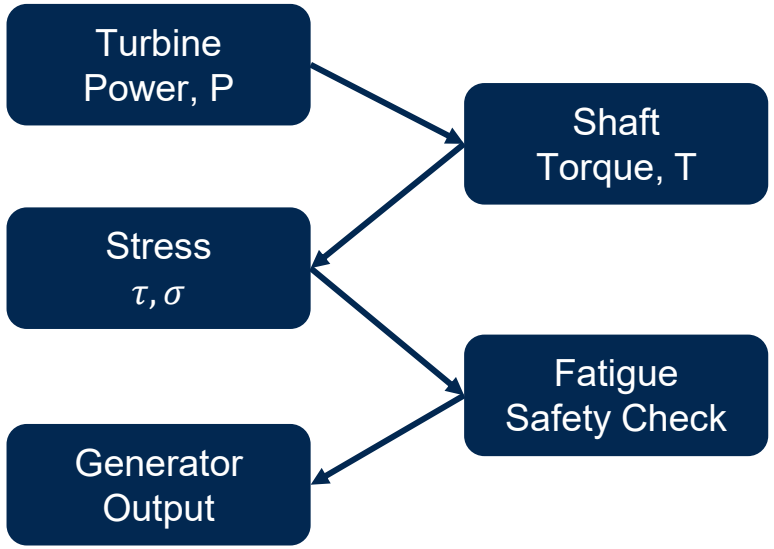
Engineering Calculations

Design Question

Design Question: What size of shaft will be needed to transmit the generated roational power into electricity by the generator in the different dams

Governing Equations

Concept	Equation	Meaning
Torque from Power	$T = \frac{P}{\omega}$	Torque on the shaft provided from the turbine and magnetic fluc from generator
Torsional Shear Stress	$\tau = \frac{Tr}{J}$	Stress on shaft from the torque
Solid Shaft Form	$\tau_{max} = \frac{16T}{\pi d^3}$	Maximum shear needed for design
Angle of Twist	$\theta = \frac{TL}{JG}$	Max elastic twist from operation
Bending	$\sigma = \frac{32M}{\pi d^3}$	Deflection based on bearing loads
Von mises	$\sigma_{vm} = \sqrt{\sigma^2 + 3\tau^2}$	Total stress on shaft



Engineering Calculations

Equation Name	Equation Format	Meaning	Fish Barrier	New Croton	Cochiti
Shaft Speed	$N = 120f/p$ $\omega = 2\pi N/60$	Direct drive at 60 Hz with 24 poles	300 rpm 31.416 rad/s	300 rpm 31.416 rad/s	300 rpm 31.416 rad/s
Torque from Power	$T = P/\omega$	Torque from mechanical shaft power	81.487 kN·m	161.065 kN·m	315.445 kN·m
Stress-Limited Diameter	$d_s = \sqrt[3]{[16T/(\pi\tau_a)]}$ $\tau_a = S_y/(\sqrt{3}\, n)$	Minimum diameter from von Mises yielding	161.44 mm	202.61 mm	253.49 mm
Twist-Limited Diameter	$d\theta = \sqrt[4]{[32TL/(\pi G\theta_a)]}$	Minimum diameter for a 5° twist limit	105.42 mm	125.00 mm	147.88 mm
Required Solid Diameter	$d = \max(d_s, d\theta)$	Stress governs the calculated minimum	161.44 mm	202.61 mm	253.49 mm
Torsional Shear Stress	$\tau = Tr/J = 16T/(\pi d^3)$	Maximum shear at required diameter	98.63 MPa	98.63 MPa	98.63 MPa
Angle of Twist	$\theta = TL/(JG)$ $J = \pi d^4/32$	Elastic twist over the 1 m shaft	0.909 °	0.724 °	0.579 °
Bending Stress	$\sigma = 32M/(\pi d^3)$	Bending neglected by setting M = 0	0 MPa	0 MPa	0 MPa
Von Mises Stress	$\sigma_{vm} = \sqrt{(\sigma^2 + 3\tau^2)}$	Equivalent stress at required diameter	170.83 MPa	170.83 MPa	170.83 MPa

Shaft power: Fish Barrier 2.56 MW, New Croton 5.06 MW, Cochiti 9.91 MW. S_y = 205 MPa, n = 1.2.
L = 1 m, G = 77 GPa, θ_a = 5°, M = 0, steady torque. τ_a = 98.63 MPa, $\sigma_{vm,allow}$ = 170.83 MPa.
Calculated static minimum diameters. Fatigue, keyways, critical speed and turbine-speed suitability require separate checks.

Engineering Calculations

Design Question

How does maintaining a minimum environmental flow affect the amount of water available for power generation?

Governing Equations

Concept	Equation	Meaning
Available Turbine Flow	$Q_{turbine} = Q_{river} - Q_{env}$	Determines flow available after required environmental considerations.
Hydraulic Power	$P = \rho g Q_{turbine} H \eta$	Calculates power generation from available turbine flow.
Power Reduction	$\Delta P = \rho g Q_{env} H \eta$	Quantifies potential power lost due to environmental flow.
Annual Energy Impact	$\Delta E = \Delta P t$	Calculates annual energy not generated due to environmental flow.
Power Availability Ratio	$R_p = 1 - Q_{env}/Q_{river}$	Represents the flow's theoretical generation potential after losses.

Variable Definitions & Sources

Variable	Source
Q_{river} & Q_{env} (River flow & Required Environmental flow)	Site Dependent on streamflow data & State environmental requirements
H (Head)	National Inventory of Dams Database
η (efficiency)	Quantifies potential power lost due to environmental flow.
P & g (Water density& Gravity acceleration)	Constants

Engineering Calculations

1. Available Turbine Flow

$$Q_{turbine} = Q_{river} - Q_{env}$$
$$Q_{turbine} = 20 - 6$$
$$Q_{turbine} = 14\ m^3/s$$

2. Hydraulic Power

$$P = \rho g Q_{turbine} H \eta$$
$$= (1000)(9.81)(14)(10)(0.80)$$
$$P = 1.099\ MW$$

3. Power Reduction

$$\Delta P = \rho g Q_{env} H \eta$$
$$= (1000)(9.81)(6)(10)(0.80)$$
$$P = 0.471\ MW$$

4. Annual Energy Impact

$$\Delta E = \Delta P t$$
$$= (0.471)(8760)$$
$$\Delta E = 4126\ MWh$$

5. Power Availability Ratio

$$R_p = 1 - Q_{env}/Q_{river}$$
$$= 1 - 6 / 20$$
$$R_p = 70\%$$

This calculation evaluates how environmental flow requirements reduce the flow available for generation, providing additional insight into site feasibility and the resulting power potential.

Engineering Calculations

- Question: What other factors impact the success of an NPD conversion?
- Expand NPD ranking from Oak Ridge National Laboratory inventory [5] to include energy loss and need

Existing

Metric	Optimal Condition	Scoring
Theoretical Yearly Power Generation	Highest yearly output (MWh)	$S_{energy} = {E_{annual} - E_{min}}/{E_{max} - E_{min}}$
Flow Variance	Lowest flow rate variance month-to-month	$S_{flow} = {CV_{flow,max} - CV_{flow}}/{CV_{flow,max} - CV_{flow,min}}$
Head Variance	Lowest head variance month-to-month	$S_{head} = {CV_{head,max} - CV_{head}}/{CV_{head,max} - CV_{head,min}}$

New

New Metric	Optimal Condition	Scoring
Energy Loss (Transmission)	Lowest distance to nearest substation	$S_{loss} = {E_{loss,max} - E_{loss}}/{E_{loss,max} - E_{loss,min}}$
Energy Need	Highest total energy need	$S_{rate} = {C_{kw} - C_{kw,min}}/{C_{kw,max} - C_{kw,min}}$

Engineering Calculations

Example Score Assessment – Cochiti Dam

Energy Output, Flow Variance, Head Variance (Existing Analysis)

$$S_{energy} = {E_{annual,max} - E_{annual}}/{E_{annual,max} - E_{annual,min}} = {58407 - 2411}/{58407 - 2411} = 1$$

$$S_{flow} = {CV_{flow,max} - CV_{flow}}/{CV_{flow,max} - CV_{flow,min}} = {1.19 - 0.61}/{1.19 - 0.02} = 0.496$$

$$S_{head} = {CV_{head,max} - CV_{head}}/{CV_{head,max} - CV_{head,min}} = {0.44 - 0.02}/{0.44 - 0} = 0.955$$

Engineering Calculations

Example Score Assessment – Cochiti Dam

Energy Loss

$$E_{loss} = \frac{P \cdot r \cdot D}{V^2} = \frac{P_{annual} \cdot r \cdot D \cdot 10^6}{V^2 \cdot 365 \cdot 24 \cdot (10^3)^2}$$

$$P_{annual} = 58407\ MW/yr\ ; D = 0.6\ mi$$

$$r = 0.3\ \Omega\ /\ mi; V = 34.5\ kV$$

$$E_{loss} = \frac{58407 \cdot 0.3 \cdot 0.6}{34.5^2 \cdot 365 \cdot 24} = 0.00106 = 0.106\%$$

$$E_{loss,max} = \frac{30000 \cdot 0.3 \cdot 12.7}{34.5^2 \cdot 365 \cdot 24} = 1.099\%$$

$$E_{loss,min} = \frac{3726 \cdot 0.3 \cdot 0.07}{34.5^2 \cdot 365 \cdot 24} = 0.00079\%$$

$$S_{loss} = {E_{loss,max} - E_{loss}}\Big/{E_{loss,max} - E_{loss,min}}$$

$$= {1.099 - 0.106}\Big/{1.099 - 0.00079} = 0.904$$

Engineering Calculations

Example Score Assessment – Cochiti Dam

Energy Need

$$E_{value} = P \cdot C_{kw} = P_{annual} \cdot 10^3 \cdot C_{kw}$$

$$P_{annual} = 58407\ MW\ /\ yr$$

$$C_{kw} = 0.134\ \$/kWh$$

$$E_{value} = 58407 \cdot 10^3 \cdot 0.134$$

$$E_{value} = \$7{,}823{,}000/yr$$

$$C_{kw,min} = 0.091\ \$/kWh\,,\ C_{kw,max} = 0.310\ \$/kWh$$

$$S_{need} = {C_{kw} - C_{kw,min}}/{C_{kw,max} - C_{kw,min}}$$

$$= {0.134 - 0.091}/{0.31 - 0.091} = 0.196$$

Engineering Calculations

Design Question

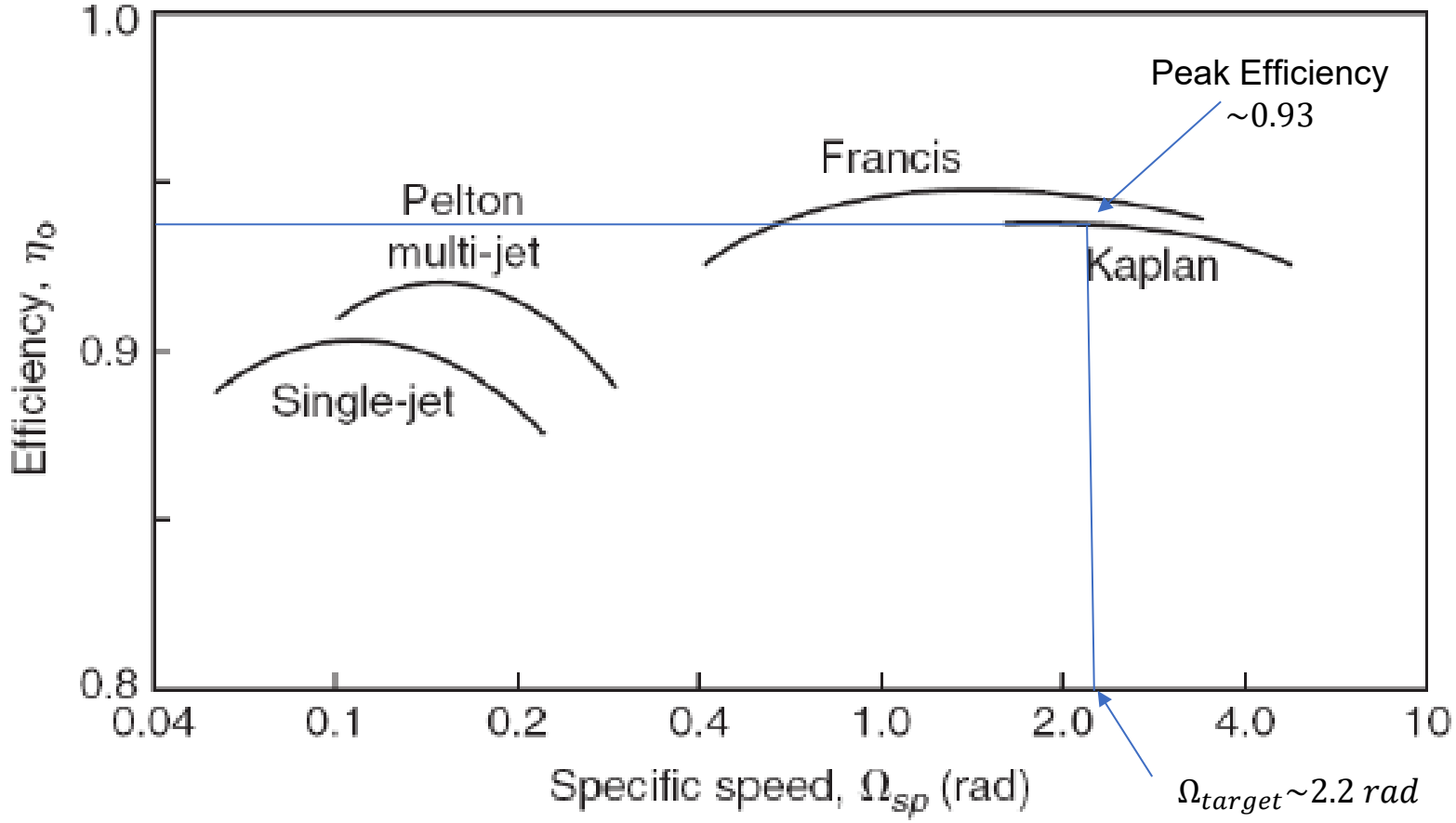
Design Question: Assume direct-drive coupling, operating at constant flow and head. Given a dam's flow and head, and the chosen turbine type, at what rotational speed is the turbine most efficient, and which generator pole counts produce those speeds?

Governing Equations

Concept	Equation	Meaning	Variable Breakdown
Target Rotational Speed	$N_{target} = \frac{60\Omega_{target}(gH_{net})^{3/4}}{2\pi\sqrt{Q_{eff}}}$	Target rotational speed of the turbine-generator shaft, based on turbine specifications	Ω_{target}: target specific speed of turbine g: gravity constant; H_{net}: hydraulic head Q_{eff}: effective flow rate
Designed Generator Pole Count	$p = \text{round}\left(\frac{120f}{N_{target}}\right)$	Pole count of generator, based on turbine specifications (up or down to even integer for north/south)	f: frequency of generator (US – 60 Hz)
Designed Rotational Speed	$N_{design} = \frac{120f}{p}$	Designed Rotational Speed, based on generator configuration	---
Angular Speed of Shaft	$\omega = \frac{2\pi N_{design}}{60}$	Angular speed of shaft, given designed rotational speed	---
Design Specific Speed	$\Omega_{\text{design}} = \frac{\omega\sqrt{Q_{eff}}}{(gH_{net})^{3/4}}$	Specific speed, given angular speed of shaft	---

Engineering Calculations

Example Assessment – Cochiti Dam (Kaplan Turbine)



Engineering Calculations

Example Assessment – Cochiti Dam (Kaplan Turbine)

$$Q_{eff} = 31.98\, m^3/s\,, H_{net} = 33.43\, m, \Omega_{target} = 2.2\, rad$$

$$N_{target} = \frac{60\Omega_{target}(gH_{net})^{3/4}}{2\pi\sqrt{Q_{eff}}} = \frac{60 \cdot 2.2 \cdot (9.81 \cdot 33.43)^{3/4}}{2\pi\sqrt{31.98}} = 286.3\, rpm$$

$$p = \text{round}\left(\frac{120 \cdot 60}{286.3}\right) = 25.1\, poles \Rightarrow p_{lower} = 24\, poles;\ \ p_{upper} = 26\, poles$$

$$N_{design,lower} = \frac{120f}{p_{lower}} = \frac{120 \cdot 60}{24} = 300\, rpm;\ \ N_{design,upper} = \frac{120f}{p_{upper}} = \frac{120 \cdot 60}{26} = 277\, rpm$$

$$\omega_{lower} = \frac{2\pi N_{design,lower}}{60} = \frac{2\pi \cdot 300}{60} = 31.4\frac{rad}{s}; \omega_{upper} = \frac{2\pi N_{design,upper}}{60} = \frac{2\pi \cdot 277}{60} = 29\frac{rad}{s}$$

Engineering Calculations

Example Assessment – Cochiti Dam (Kaplan Turbine)

$$\Omega_{design,lower} = \frac{\omega_{lower}\sqrt{Q_{eff}}}{(gH_{net})^{3/4}} = \frac{31.4 \cdot \sqrt{31.98}}{(9.81 \cdot 33.43)^{3/4}} = 2.3\, rad$$

$$\Omega_{design,upper} = \frac{\omega_{upper}\sqrt{Q_{eff}}}{(gH_{net})^{3/4}} = \frac{29 \cdot \sqrt{31.98}}{(9.81 \cdot 33.43)^{3/4}} = 2.1\, rad$$

Both specific speeds (Ω = 2.1 and 2.3 rad) fall in the near-peak region of the Kaplan efficiency curve (~93%). Suitable generators:

- 24 poles (300 rpm) or 26 poles (277 rpm)

Pairwise Matrix – Iteration 1

Weights for MATLAB code							
	Variance of Head	Variance of Flow	Infrastructure Distance	Power Generation	Local Energy Cost	Geometric Mean	Weights
Variance of Head	1.000	2.000	1.500	0.800	1.000	1.191	0.225
Variance of Flow	0.500	1.000	0.300	0.330	0.500	0.477	0.090
Infrastructure Distance	0.667	3.333	1.000	1.500	1.200	1.320	0.249
Power Generation	1.250	3.030	0.667	1.000	0.800	1.151	0.217
Local Energy Cost	1.000	2.000	0.833	1.250	1.000	1.158	0.219

Decision Matrix – Dam and Turbine

Criterion	Weight	New Croton Reservoir Dam		Little Falls Dam		Birch Hill Dam	
		Score out of 1	Weighted Score	Score out of 1	Weighted Score	Score out of 1	Weighted Score
Estimated Power Output	21.7%	0.47	0.102	0.72	0.156	0.09	0.020
Flow Rate Variance	9.0%	0.47	0.042	0.46	0.041	0.49	0.044
Head Variance	22.5%	0.92	0.207	0.92	0.207	0.88	0.198
Energy Loss (Transmission)	24.9%	0.99	0.247	0.92	0.229	0.96	0.239
Energy Need	21.9%	0.90	0.197	0.29	0.064	0.89	0.195
Total	100%		0.79		0.70		0.70
Rank		1		2		3	

Rank	Site	Optimal Turbine
1	Fish Barrier Dam	Francis
2	New Croton Reservoir Dam	Francis
3	Cochiti Dam	Kaplan

Penstock Gate Concept Evaluation

Evaluation	Fish Barrier	New Croton Reservoir	Cochiti
Design basis D, Q, H and velocity	D = 2.7 m Q = 15.57 m³/s H = 18.23 m V = 2.72 m/s	D = 2.2 m Q = 11.48 m³/s H = 34.84 m V = 3.02 m/s	D = 3.7 m Q = 31.98 m³/s H = 33.43 m V = 2.97 m/s
Fixed-wheel vertical lift gate	PRIMARY CANDIDATE Rolling supports reduce friction. Maintain wheels, tracks and seals.	PRIMARY CANDIDATE Reduced friction helps operation under hydraulic loading.	PRIMARY CANDIDATE Rolling supports suit the largest opening and highest flow.
Caterpillar / tractor gate	OTHER ALTERNATIVE Roller chains add maintenance without an established benefit.	OTHER ALTERNATIVE Consider if distributed support improves the gate design.	SECONDARY CANDIDATE Distributed rollers may help carry the larger gate load.
High-pressure slide gate	SECONDARY CANDIDATE Simple support arrangement. Verify friction and actuator force.	SECONDARY CANDIDATE Smallest opening favors evaluation. Check pressure-dependent friction.	LOWER PRIORITY Larger opening increases concern about sliding friction and actuation.
Primary choice	**Fixed-wheel gate**	**Fixed-wheel gate**	**Fixed-wheel gate**
Secondary candidate	**High-pressure slide gate**	**High-pressure slide gate**	**Caterpillar / tractor gate**

Preliminary engineering judgment. All candidates require rated closure under maximum flow and head, with a defined leakage limit.
H is the earlier turbine head, not a confirmed gate differential head. Final selection needs gate geometry, closure time and transient/air-vent checks.
Secondary means an alternative concept for evaluation, not an additional installed backup gate.

Fish passage Concept Evaluation

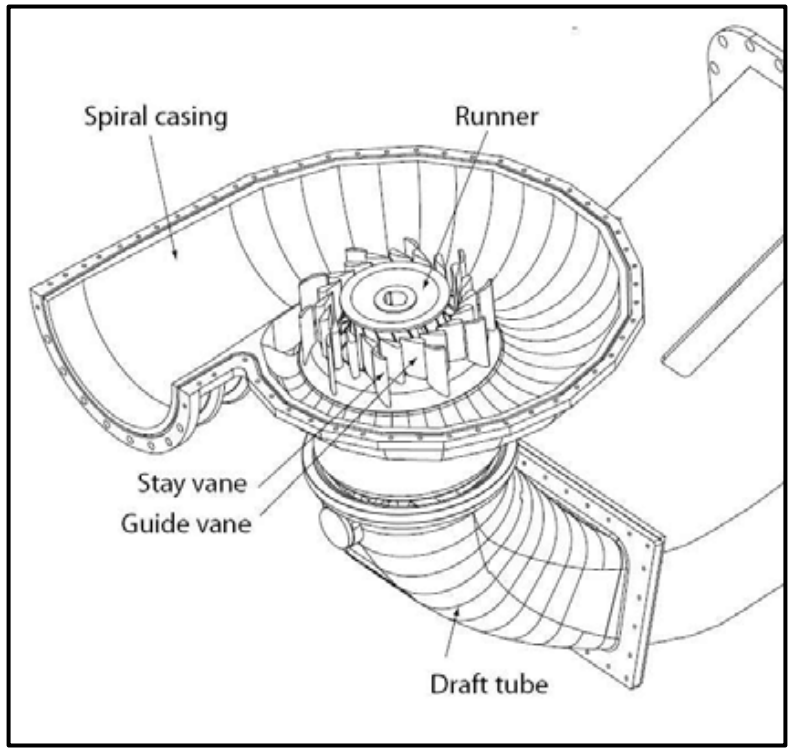
- Dam height favors a lift: all three candidate sites have heads of about 60 to 114 ft, well above the roughly 25 ft height where fish lifts are generally preferred [20].
- Other options are long and expensive: a fish ladder would need a long series of concrete pools to climb the dam, and a nature-like bypass would need a low-slope channel stretching a great distance around it [20].
- A lift is simpler and cheaper: it carries fish over the dam in a single trip, and its structure costs less than the large amount of concrete a tall ladder requires [20].

BEST CHOICE: LIFT

Concept Selection – Dam and Turbine

Fish Barrier Dam & Francis Turbine

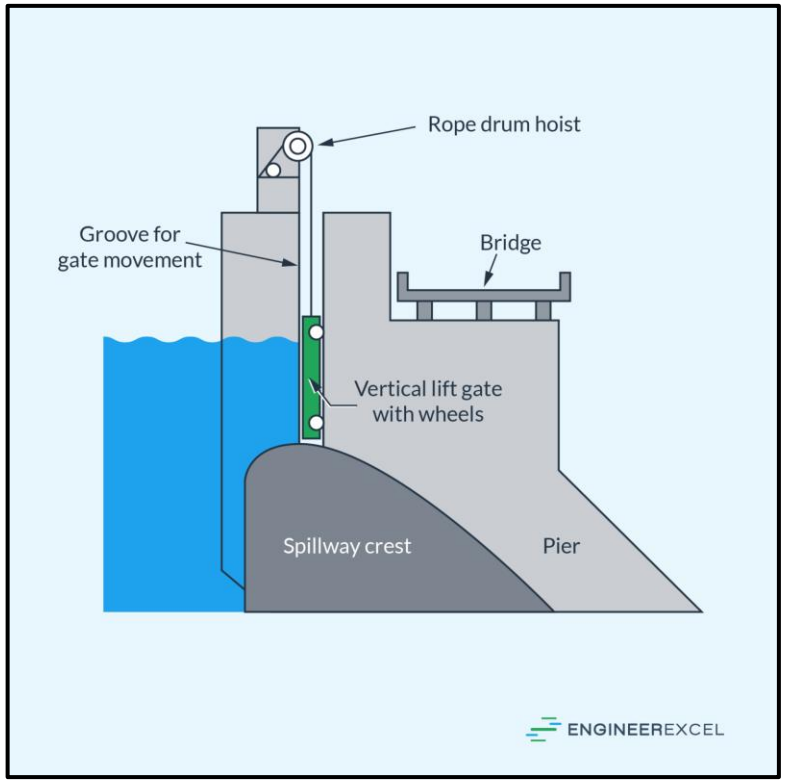




- Low variance in Head and Flow
- Minimal loss in energy during transmission
- Highest cost of energy per kWh of all dams

Concept Selection – Penstock Gates

Fixed Wheel Gate

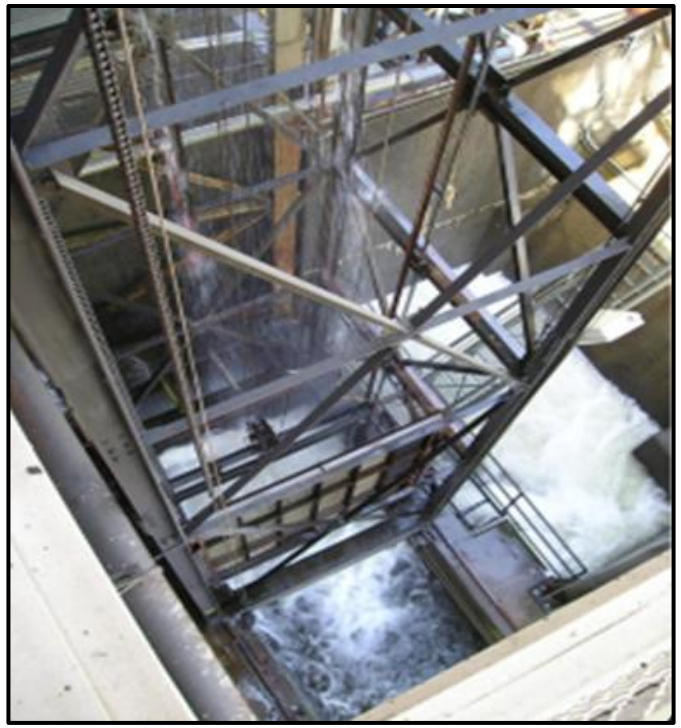




A relatively cheap, low maintenance and reliable option that is well tested in the field and can be shut in emergency situations without power. For the dams discussed in this presentation this will be our first consideration.

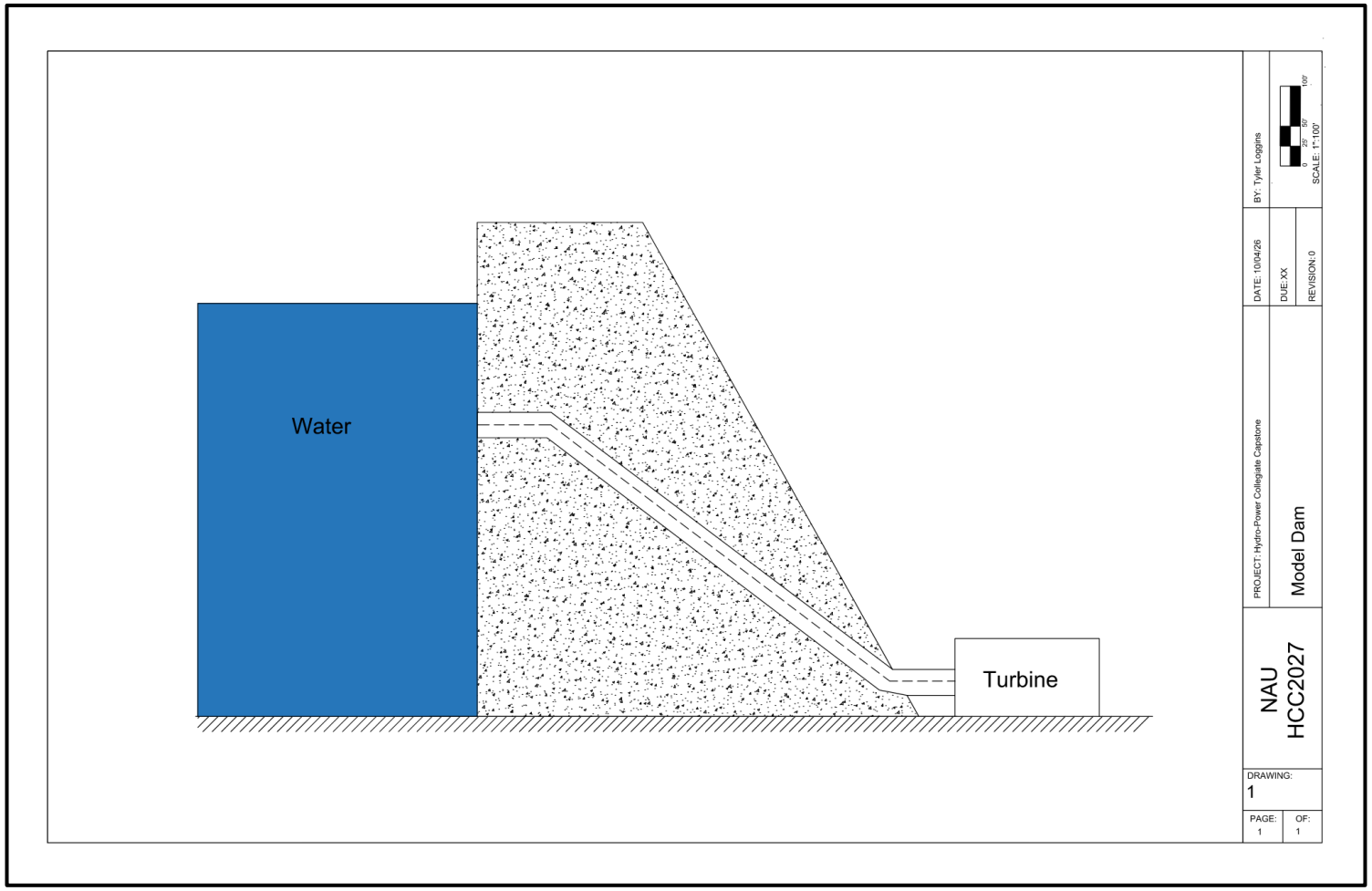
Concept Selection – Fish Passage

Fish Lift

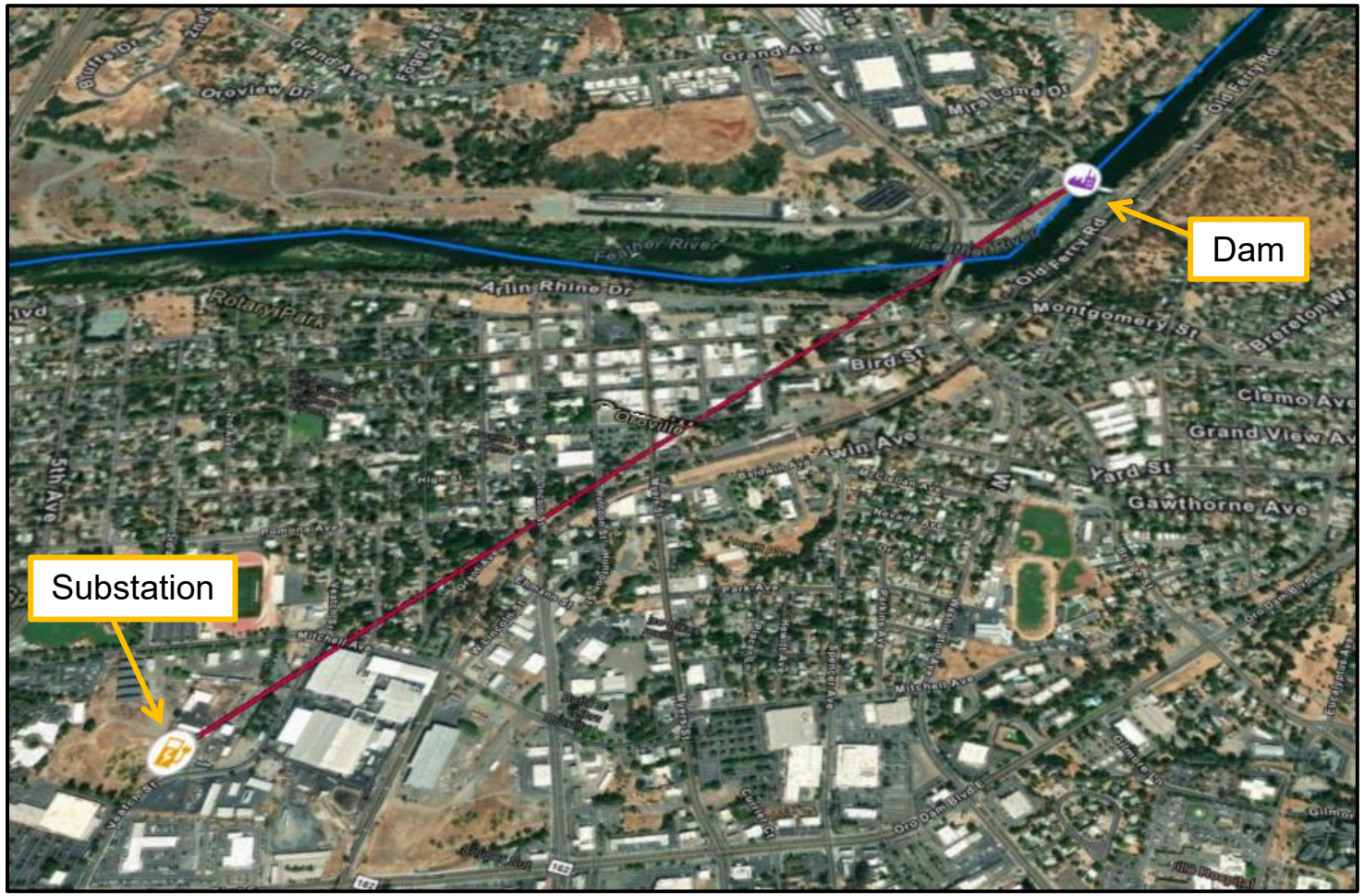


A fish lift was selected for Fish Barrier Dam because, at a 59.8 ft head, the dam exceeds the roughly 25 ft height above which lifts are generally preferred, avoiding a long ladder and costing less than the large concrete structure a ladder of that height would require.

Concept Selection – AutoCAD Rendering



Concept Selection – ArcGIS Site Rendering



From ArcGIS Pro:

- Perform advanced analysis to identify nearby transmission lines to determine feasibility
- Consider impacts of dam and transmission lines on surrounding residential areas (soft elimination)

Budget (actual)

NAU HCC Capstone — Budget Plan		
Capstone budget	$15,000	*Given*
Fundraising goal	$1,500	*Target to be raised; not yet secured*
Total available funds	**$16,500**	

Category	Amount ($)	% of Funds	Details
Trip 1: Denver Co (Oct 22–25)	$4,645	28.2%	*Airfare, 3 rooms × 3 nights, dinners, 1 rental van*
Trip 2: Duluth Mn (May 10, 3 nights)	$9,492	57.5%	*10 travelers, 5 rooms × 3 nights, meals, 2 rental vans*
Community outreach (STEM kits)	$1,500	9.1%	*Hydropower STEM kits for local outreach*
TOTAL PLANNED	**$15,637**	**94.8%**	
Remaining (Over) Budget	**$637**	**5.2%**	**Over budget**
Remaining without fundraising	($637)		

Airfare: Estimates are based on current fares listed on Google Flights. Flights have not yet been booked, and final costs may vary with booking date and availability.[5]
Lodging: Estimates are based on rates for hotels identified near each destination. Rates have not yet been verified or reserved and are subject to change.
Meals: Meal costs are estimated at a flat rate of $25 per meal, including tax and tip.
Fundraising: The $1,500 fundraising goal will be pursued in partnership with the NAU Energy Club. These funds are projected and not yet secured.

Siting Budget (Theoretical)

Line Item	Share of Phase	Cost ($)	Basis	Cost with 2% Inflation ($)
1. Siting (average of 4 real feasibility studies)				
Hydrology, flow & head analysis	35.0%	$10,028	*Flow duration curves, head, power estimate*	$10,028
Site visit & field reconnaissance	20.0%	$5,730	*Inspect dam, access, and grid connection*	$5,730
Environmental & regulatory screening	20.0%	$5,730	*dam owner & permitting path*	$5,730
Cost estimate, economics & report	25.0%	$7,163	*Capital cost, energy*	$7,163
Siting subtotal	**100.0%**	**$28,650**		**$28,650**
2. FERC licensing (application & required studies)		$387,992	*Average of 11 original NPD licenses, 1-10 MW*	$409,818
Subtotal		**$416,642**		**$438,468**
Contingency		$83,328		$87,694
TOTAL		**$499,970**		**$526,162**

Estimates based off previous costs of NPD conversion projects [7],[9],[10],[11].

Bill of Materials (Theoretical)

Bill of Materials

Item	Qty	Unit	Material Cost (2026 $)	Specification / What's Included
Francis turbine-generator package	1	lot	$2,895,532	*Runner & wicket gates, draft tube liner, generator, hydraulic power unit, switchgear, controls & protection*
Powerhouse structure	1	lot	$1,050,293	*Concrete, rebar, structural steel, roofing, crane rail*
Intake structure	1	lot	$315,088	*Trash rack, intake gate, stoplogs, concrete*
Penstock & water conveyance	1	lot	$266,463	*Steel penstock pipe, shutoff valve, tailrace lining*
Site access & development	1	lot	$313,143	*Access road base, fencing, drainage, erosion control*
Transformer, switchyard & grid line	1	lot	$1,120,312	*Step-up transformer, breakers, metering, poles & conductor to grid*
TOTAL MATERIALS			**$5,960,831**	

Estimates made from head and turbine estimates [2],[4].

Construction Budget (Theoretical)

Category	Cost (2026 $)	% of Total	Cost with 2% Inflation ($)	Basis
Materials (equipment & construction materials)	$5,960,831	38.3%	$6,779,992	*Bill of Materials tab*
Labor (installation, construction, engineering)	$9,599,061	61.7%	$10,918,202	*Labor & Installation section*
TOTAL	**$15,559,891**	**100.0%**	**$17,698,194**	***2-year construction 2032–2033***

Estimates made from head and turbine estimates [2],[4].

Schedule

WBS	Task / Deliverable	Start Date	End Date	October				November					December				January				February				Status
				W7	W8	W9	W10	W11	W12	W13	W14	W15	W16	W17	W18	W19	W20	W21	W22	W23	W24	W25	W26	W27	
				10/5	10/12	10/19	10/26	11/2	11/9	11/16	11/23	11/30	12/7	12/14	12/21	12/28	1/4	1/11	1/18	1/25	2/1	2/8	2/15	2/22	
1.0	Project Definition & Team Planning	8/24/26	9/14/26																						Complete
2.0	HCC Research & Benchmarking	8/31/26	10/19/26																						On Schedule
3.0	Customer / Engineering Requirements & QFD	9/7/26	10/5/26																						On Schedule
4.0	NPD Site Screening & Selection	9/14/26	10/5/26																						In Progress
5.0	Literature Review	9/21/26	11/2/26																						In Progress
6.0	Mathematical / Hydropower Modeling	9/28/26	11/9/26																						In Progress
7.0	Concept Generation	10/12/26	11/2/26																						Upcoming
8.0	Decision Matrix & Pugh Chart	10/19/26	11/9/26																						Upcoming
9.0	Preliminary Engineering Design	10/26/26	11/30/26																						Upcoming
10.0	Prototype Planning / Procurement	10/26/26	11/30/26																						Upcoming
11.0	Community / Industry Engagement	9/14/26	2/22/27																						On Schedule
12.0	ME 476 Fall Deliverables	10/12/26	12/7/26																						On Schedule
13.0	Siting Challenge Package	12/7/26	1/25/27																						Upcoming
14.0	Design Challenge Package	1/11/27	2/22/27																						Upcoming
15.0	Community Connections Package	1/11/27	2/22/27																						Upcoming
16.0	Optional Build & Test Plan	1/11/27	2/22/27																						Upcoming
17.0	Final Event Prep (Poster / Quick Pitch / Final Report)	2/23/27	5/6/27																						Upcoming

Winter Break

Future Work

Siting Challenge

- Implement further elimination criteria (protected land consideration, fish passage feasibility)
- Quantify and verify current “soft elimination” criteria (transmission through elevation changes, residential areas)
- Add onto and improve upon current selection criteria (cost to produce energy, MW capacity per month, construction costs)

Design Challenge

- Improve on current AutoCAD model
- Investigate more into turbine efficiency and implementation considerations
- Investigate conversion considerations for electrical infrastructure (generator)

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Thank You
& Questions?

Customer Requirements - Siting

Requirements	Decided weights	Our understanding
Resource and generation ability	5	The abundance of the water allowing us to generate power
Environmental factors	4	Fish, acres flooded, downstream distrubance, habitat change
Dam safety	3	Could this alteration cause future problems for the dam
Access to transmission and grid integration	5	Ease of access to existing grids and power stations
Transportation access	2	Ease of access to the area for installation
Cultural factors	1	the existing dam/lake/river important or its surrounding/inhabitants affected
Operation and maintenance requirements	3	Self explanatory
Agricultural effects	3	The use of water from this dam could affect farms downstream or pulling from resevoir
Social metrics	2	Will the local people like or dislike the addition, rec area, bill decrease, noise polution
High level costs	5	Installation costs of the turbine facility, access roads, high voltage cables, drilling hole in dam safely,

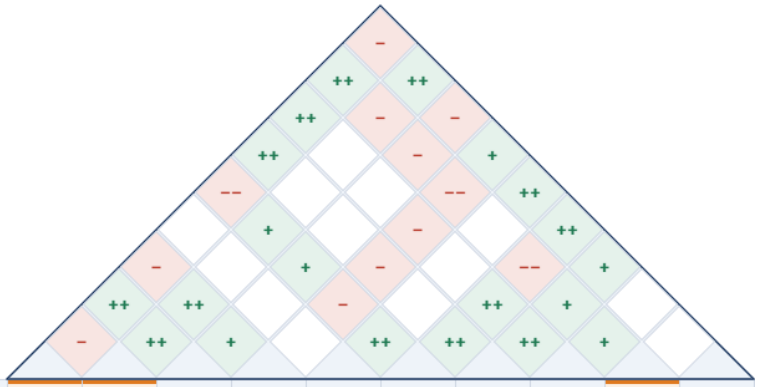
Engineering Requirements - Siting

EGR REQ	QFD rank	Goal
Instalation Cost	1	$8-10 million
Max Head	2	30-50 m
Required Flow	7	24-60 m^3/s
Resevoir Capacity	5	1-10 million m^3
Local Electric Capacity	9	>13 cents/kWh
Cost Recup Time	4	1 year
Infrastructure Distance	5	1 mile
Site Accesibility	10	10 miles
Ecological Impact	2	<15000 acres disturbed, among other wildlife and habitat considerations
Energy production	8	35,000 MWh/yr

QFD

Relationship (grid)
● Strong 9 ○ Medium 3 ▽ Weak 1

Correlation (roof)
++ Strong synergy + Synergy
− Trade-off −− Strong trade-off



Customer needs What customers want, in their own words	Importance (1 to 5)	instalation cost	max head	required flow	resevoir capacity	local electric prices	cost recup time	infrastructure distance	site accesibility	ecological impact	energy production
Direction of improvement		▼	▲	▼	▲	▲	▼	▼	▲	▼	◎
resource and generation ability	5	○	●	●	●	●	●	○	▽	●	●
enviromental factors	4	▽	○	○	●	●	▽	○	○	●	○
dam safety	3	○	●	○	●					○	
access to transmission and grid integ	5	●				○	●	●	○	▽	▽
transportation access	2	●					○	○	●	○	
cultural factors	1	▽	▽	●	○			▽	○	●	
operation and maintanance requirem	3	○	○	●	▽		●	▽	○		○
agricultural effects	3	▽	▽	○	○			▽	▽	●	▽
social metrics (will the people approv	2	○	○	○	▽	○			○	○	▽
high level costs	5	●	●	▽	▽	○	○	●	○	▽	●

Target value	8-10 million	50M	24-60 M^3/s	1-10 million m^3	>13 cents/kWh	1 year	1 mile	10 miles	<15000 acres disturbed	35,000 MWh/yr
Importance score	155	148	122	130	117	142	130	86	148	121
Relative weight	12%	11%	9%	10%	9%	11%	10%	7%	11%	9%
Priority rank	#1	#2	#7	#5	#9	#4	#5	#10	#2	#8

Pairwise Matrix – Iteration 1

Weights for MATLAB code							
	Variance of Head	Variance of Flow	Infrastructure Distance	Power Generation	Local Energy Cost	Geometric Mean	Weights
Variance of Head	1.000	2.000	1.500	0.800	1.000	1.191	0.225
Variance of Flow	0.500	1.000	0.300	0.330	0.500	0.477	0.090
Infrastructure Distance	0.667	3.333	1.000	1.500	1.200	1.320	0.249
Power Generation	1.250	3.030	0.667	1.000	0.800	1.151	0.217
Local Energy Cost	1.000	2.000	0.833	1.250	1.000	1.158	0.219

Decision Matrix

#	Dam	St	River	MW	MWh/yr	Q cfs	Head...	Sub mi	$/kWh	Q CV	H CV	Score
1	FISH BARRIER	CA	Feather River	2.5600	19868	549.7000	59.8000	1.3000	0.3098	0.0200	0.0400	0.8120
2	NEW CROTON RESERVOIR DAM	NY	Croton River	5.0600	20192	405.3000	114.3000	0.2000	0.2876	0.6400	0.0400	0.7550
3	ANDERSON COTTONWOOD	CA	Sacramento Rever	7.4400	39618	7.7498e+	8.7000	0.2000	0.2379	0.4200	0.1300	0.7490
4	COCHITI DAM	NM	Rio Grande & Santa Fe	9.9100	58407	1.1294e+	109.7000	0.6000	0.1340	0.6100	0.0200	0.7410
5	BIRCH HILL DAM	MA	Millers River	1.2100	5818	611.9000	20.2000	2.6000	0.2853	0.6200	0.0500	0.6810
6	PALO VERDE DIVERSION	AZ	Colorado River	3.8600	22225	1.2377e+	28.6000	0.4000	0.2011	0.3500	0.0800	0.6700
7	FISH BARRIER DAM	WA	Baker River	7.8800	40651	2.5879e+	25.8000	0.2000	0.1097	0.2500	0.0700	0.6660
8	LOCK E10 CRANESVILLE	NY	Mohawk River (Erie Canal)	5.4600	25803	5.3003e+	10.7000	0.2000	0.1864	0.5900	0.0700	0.6600
9	NORTH FORK DAM	CA	North Fork American River	3.7200	10289	552	55.9000	1.6000	0.3098	0.9500	0.1300	0.6590
10	LAST CHANCE WEIR	CA	Kings River	1.2300	4438	1.1387e+	8.7000	1.5000	0.3098	0.7500	0.1300	0.6570

Construction Validation

1. Site Specifications
2. Engineering Analysis
3. Component Sizing
4. Structural Checks
5. Final Design

Construction Variable	Site Dependent Factors	Planned Validation
Turbine Size & Type	Head, Flow, Available space	Hydraulic Analysis & Manufacturer Data
Structural Support Addition	Existing Structure, equipment loads, attachment location	Load Calculations & Structural FEA
Foundation Modifications	Existing foundation + soil conditions	Structural Analysis & Site Investigation
Gates & Valves	Flow, pressure, existing infrastructure	Load Calculations, Manufacturer specifications
Fish Passage Considerations	Debris conditions, Fish Requirements, Flow	Hydraulic Analysis & Fish Passage requirements
Equipment Access	Site layout, equipment dimensions, construction access	Site surveying, installation planning
Electrical Integration	Generator output, existing infrastructure	Electrical load, Connection Analysis

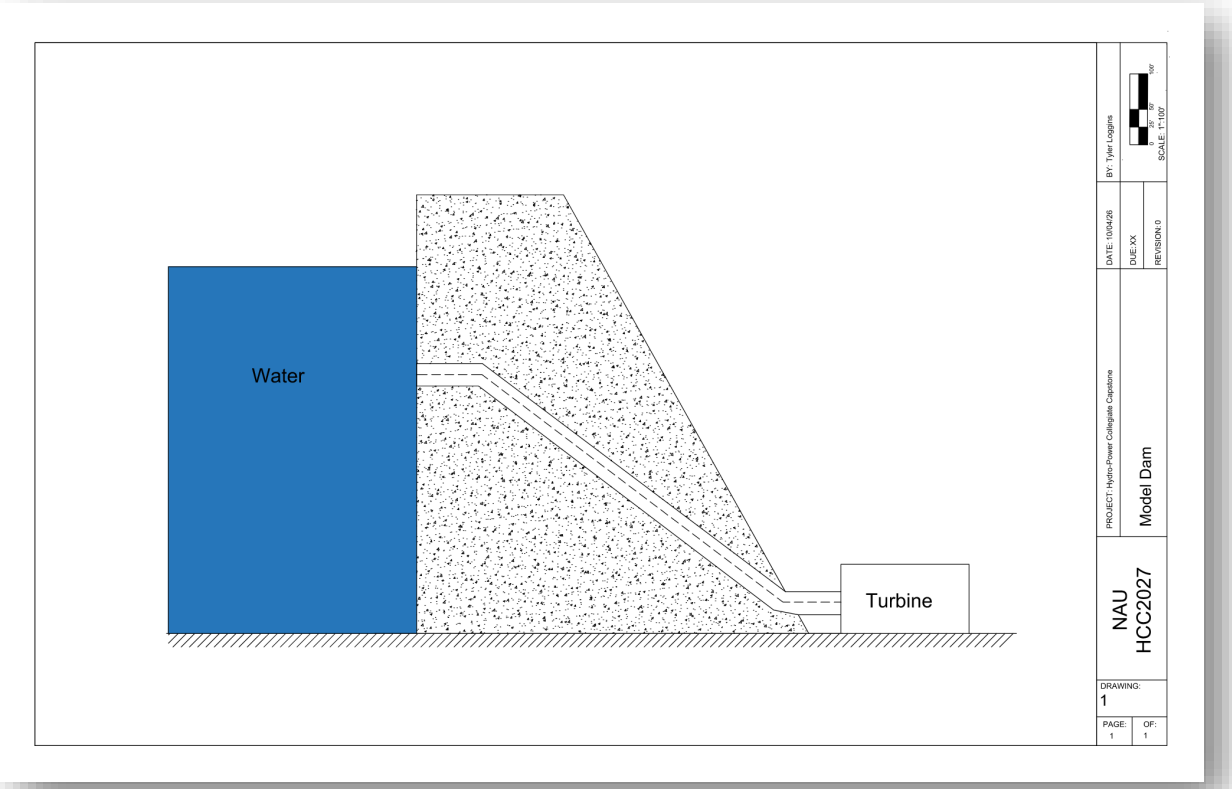
Mathematical Modelling

Design Question

How does penstock diameter affect water velocity and hydraulic head loss ? What diameter should be carried forward for preliminary design?

Governing Equations

Concept	Equation	Meaning
Area	$A = \frac{\pi D^2}{4}$	Cross-sectional area of penstock
Velocity	$V = \frac{Q}{A}$	Average water velocity in penstock
Friction Loss (Darcy-Weisbach)	$h_f = f \frac{L}{D} \frac{V^2}{2g}$	Head loss due to pipe friction
Minor Losses	$h_m = \sum K \frac{V^2}{2g}$	Head loss from fittings, entrances, bends, ect.
Total Head Loss	$h_L = h_f + h_m$	Total hydraulic losses in penstock



Mathematical Modelling

Governing Equation

$$Q_n = \left[\frac{\gamma\, q_{50}}{1+(\gamma-1)\, q_{50}} \times \left(1-\frac{q_{min}}{q_{max}}\right)+\frac{q_{min}}{q_{max}}\right] Q_{max}$$

Validation Example

Verde River annual mean flows (USGS 09506000, 1996–2025) used to validate the optimum-flow equation.

Key Takeaways

- 30-year Verde River record gives Q_n = 207.3 ft³/s for a Francis turbine.
- The method uses flow-duration data to select the design flow.
- Helps balance energy capture and project cost.
- Can be applied to candidate sites and turbine options for comparison.

Calculations

Parameter	Calculation / Value	Result
Maximum flow rate (1996–2025)	$Q_{max} = 1133.6\ ft^3/s$	1133.6 ft³/s
Normalized median flow	$q_{50} = \frac{Q_{50}}{Q_{max}} = \frac{214.2}{1133.6}$	0.189
Normalized minimum flow	$q_{min} = \frac{Q_{min}}{Q_{max}} = \frac{116.2}{1133.6}$	0.102
Normalized maximum flow	$q_{max} = 1$	1
Turbine coefficient (Francis)	$\gamma = 0.422$ (Francis)	0.422
Optimum design flow	$Q_n = 207.3\ ft^3/s$	**207.3 ft³/s**

Mathematical Modelling

Design Question

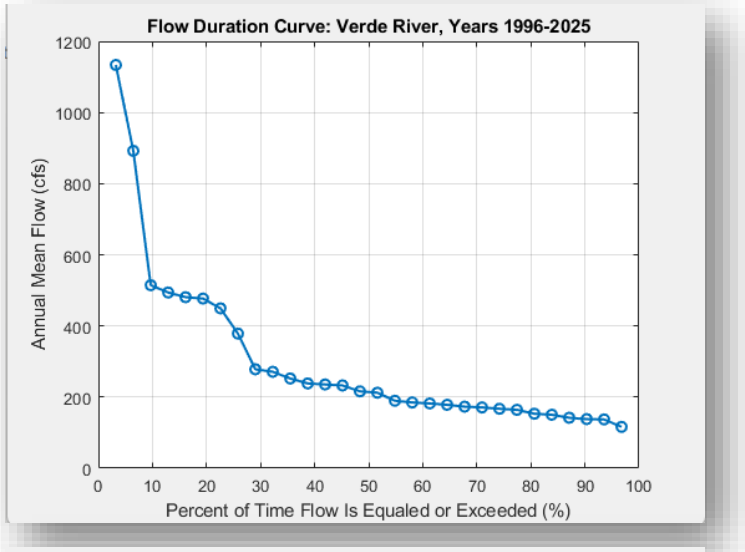
How often does the river have enough water to run a turbine?

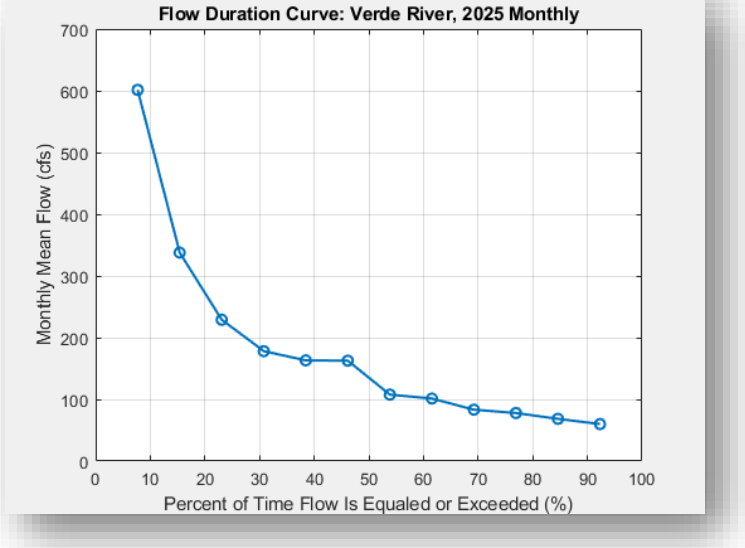
Approach

Make a flow duration curve, a chart showing how often each flow level happens.

Application

Use the same method on our chosen sites to design the turbine and other parts.

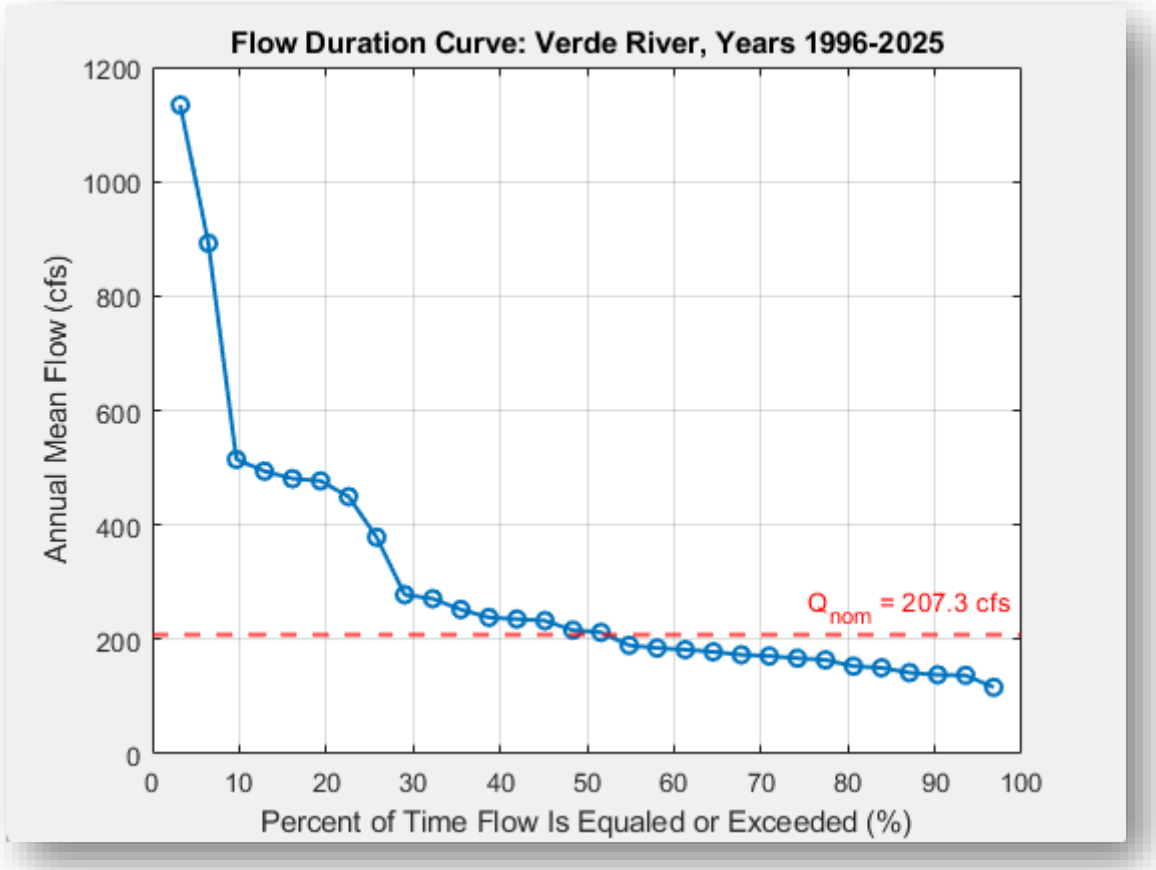




Mathematical Modelling

Results

This chart shows that the optimum design flow of 207.3 CFS, which balances power output and cost, was met or exceeded about 52% of the time over the last 30 years.



Mathematical Modelling

$$E_{loss} = \frac{P \cdot r \cdot D}{V^2} \qquad E_{value} = P \cdot C_{kw}$$

- Question: What other factors impact the success of an NPD conversion?
- Expand NPD ranking from Oak Ridge National Laboratory inventory [5] to include energy loss and need

Current

Metric	Optimal Condition	Scoring
Theoretical Yearly Power Generation	Highest yearly output (MWh)	$S_{energy} = {E_{annual} - E_{min}}/{E_{max} - E_{min}}$
Flow Variance	Lowest flow rate variance month-to-month	$S_{flow} = {CV_{flow,max} - CV_{flow}}/{CV_{flow,max} - CV_{flow,min}}$
Head Variance	Lowest head variance month-to-month	$S_{head} = {CV_{head,max} - CV_{head}}/{CV_{head,max} - CV_{head,min}}$

New

New Metric	Optimal Condition	Scoring
Energy Loss (Transmission)	Lowest distance to nearest substation	$S_{loss} = {E_{loss,max} - E_{loss}}/{E_{loss,max} - E_{loss,min}}$
Energy Need	Highest total energy need	$S_{rate} = {C_{kw} - C_{kw,min}}/{C_{kw,max} - C_{kw,min}}$

Mathematical Modelling

Example Score Assessment – Cochiti Dam

Energy Loss

$$E_{loss} = \frac{P \cdot r \cdot D}{V^2} = \frac{P_{annual} \cdot r \cdot D \cdot 10^6}{V^2 \cdot 365 \cdot 24 \cdot (10^3)^2}$$

$$P_{annual} = 58407\ MW/yr\ ; D = 0.6\ mi$$

$$r = 0.3\ \Omega\ /\ mi; V = 34.5\ kV$$

$$E_{loss} = \frac{58407 \cdot 0.3 \cdot 0.6}{34.5^2 \cdot 365 \cdot 24} = 0.00106 = 0.106\%$$

$$E_{loss,max} = \frac{30000 \cdot 0.3 \cdot 12.7}{34.5^2 \cdot 365 \cdot 24} = 1.099\%$$

$$E_{loss,min} = \frac{3726 \cdot 0.3 \cdot 0.07}{34.5^2 \cdot 365 \cdot 24} = 0.00079\%$$

$$S_{loss} = {E_{loss,max} - E_{loss}}\Big/{E_{loss,max} - E_{loss,min}}$$

$$= {1.099 - 0.106}\Big/{1.099 - 0.00079} = 0.904$$

Mathematical Modelling

Example Score Assessment – Cochiti Dam

Energy Need

$$E_{value} = P \cdot C_{kw} = P_{annual} \cdot 10^3 \cdot C_{kw}$$

$$P_{annual} = 58407\ MW\ /\ yr$$

$$C_{kw} = 0.134\ \$/kw$$

$$E_{value} = 58407 \cdot 10^3 \cdot 0.134$$

$$E_{value} = \$7{,}823{,}000/yr$$

$$C_{kw,min} = 0.091\ \$/kw\ ,\ C_{kw,max} = 0.310\ \$/kw$$

$$S_{need} = {C_{kw} - C_{kw,min}}\Big/{C_{kw,max} - C_{kw,min}}$$

$$= {0.134 - 0.091}/{0.31 - 0.091} = 0.196$$

Mathematical Modelling

1. Available Turbine Flow

$$Q_{turbine} = Q_{river} - Q_{env}$$
$$Q_{turbine} = 20 - 6$$
$$Q_{turbine} = 14\ m^3/s$$

2. Hydraulic Power

$$P = \rho g Q_{turbine} H \eta$$
$$= (1000)(9.81)(14)(10)(0.80)$$
$$P = 1.099\ MW$$

3. Power Reduction

$$\Delta P = \rho g Q_{env} H \eta$$
$$= (1000)(9.81)(6)(10)(0.80)$$
$$P = 0.471\ MW$$

4. Annual Energy Impact

$$\Delta E = \Delta P t$$
$$= (0.471)(8760)$$
$$\Delta E = 4126\ MWh$$

5. Power Availability Ratio

$$R_p = 1 - Q_{env}/Q_{river}$$
$$= 1 - 6 / 20$$
$$R_p = 70\%$$

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Thank You
& Questions?

Decision Matrix

#	Dam	State	River	MW	MWh/yr	Sub mi	$/kWh	Score
1	Fish Barrier	CA	Feather River	2.56	19868	1.3	0.3098	0.8160
2	NewCorton Reservoir Dam	NY	Croton River	5.06	20192	0.2	0.2876	0.7630
3	Anderson Cottonwood	CA	Sacramento Rever	7.44	39618	0.2	0.2379	0.7570
4	Cochiti Dam	NM	Rio Grande & Santa Fe	9.91	58407	0.6	0.1340	0.7460
5	Birch Hill Dam	MA	Millers River	1.21	5818	2.6	0.2853	0.6900
6	Palo Verde Diversion	AZ	Colorado River	3.86	22225	0.4	0.2011	0.6800
7	Fish Barrier Dam	WA	Baker River	7.88	40651	0.2	0.1097	0.6760
8	Lock E10 Cranesville	NY	Mohawk River (Erie Canal)	5.46	25803	0.2	0.1864	0.6700
9	North Fork Dam	CA	North Fork American River	3.72	10289	1.6	0.3089	0.6680
10	Last Chance Weir	CA	Kings River	1.23	4438	1.5	0.3089	0.6670